What Caused the Accident? Speculating Accessibility Challenges in Automated Buses for Older Adults

Wanjun Chu
Department of Management and Engineering
Linköping University
Linköping, Sweden
wanjun.chu@liu.se

Torbjörn Andersson
Department of Management and Engineering
Linköping University
Linköping, Sweden
torbjorn.andersson@liu.se

Fredrik Henriksson
Department of Management and Engineering
Linköping University
Linköping, Sweden
fredrik.henriksson@liu.se

Simon Schütte
Department of Management and Engineering
Linköping University
Linköping, Sweden
simon.schutte@liu.se

Elena Jiménez Romanillos
Department of Management and Engineering
Linköping University
Linköping, Sweden
elena.jimenez.romanillos@liu.se

Franziska Felicitas Babel
Dept.of Computer and Information Science
Linköping University
Linköping, Sweden
franziska.babel@liu.se

Abstract

Automated buses are increasingly being deployed in public transport systems. In Linköping, Sweden, they have been in daily operation since 2020. However, their deployment has also raised critical accessibility and inclusive design challenges. There is growing concern regarding how automated buses can accommodate users with diverse mobility, sensory, and cognitive capabilities, especially older adults and people with impairments, throughout the entire user journey. In this workshop, we apply a speculative design approach to engage participants in investigating a fictive accident involving older adult passengers. We translate these speculative scenarios into design research questions addressing safety, accessibility, and inclusivity. Using the automated bus deployment in Linköping as the setting, the workshop aims to generate insights that bridge the gap between research and real-world deployment for a more accessible and inclusive autonomous public transport system.

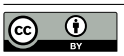
CCS Concepts

• **Human-centered computing**; • **Accessibility**; • **Accessibility systems and tools**;

Keywords

Autonomous public transportation, automated bus, accessibility, inclusive design, speculative design

ACM Reference Format:
Wanjun Chu, Torbjörn Andersson, Fredrik Henriksson, Simon Schütte, Elena Jiménez Romanillos, and Franziska Felicitas Babel. 2026. What Caused the Accident? Speculating Accessibility Challenges in Automated Buses for Older Adults. In *18th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI Adjunct '26), September 20–23, 2026, Gothenburg, Sweden.* ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/3828158.3834798



This work is licensed under a Creative Commons Attribution 4.0 International License.
AutomotiveUI Adjunct '26, Gothenburg, Sweden
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2815-0/2026/09
https://doi.org/10.1145/3828158.3834798

1 Introduction

In the city of Linköping, Sweden, automated buses have been touring the streets since 2020. The buses serve both as a testbed for autonomous public transportation and as a fully operational platform that has transported over 20,000 passengers to date. Currently, two automated buses provide services between Campus Valla at Linköping University (LiU) and a neighboring residential area as part of the *Ride the Future* project (https://ridethefuture.se/). *Ride the Future* is co-funded by the Swedish Agency for Economic and Regional Growth (Tillväxtverket) and the European Union and brings together partners across civil society, academia, and the private sector. The project is currently transitioning into a new phase in which two next-generation automated buses will replace the existing ones. They are expected to offer increased operational range, higher passenger capacity, adaptable design configurations, and higher speeds. The operational area is also expected to expand into additional residential and shopping areas.

This transition creates a unique window of opportunity to engage more users with autonomous public transportation and learn from real-world deployment in everyday contexts. However, it also raises critical accessibility and inclusive design challenges. We see increasing demand from the research community, industry, and society at large to consider the needs and requirements of users with diverse mobility, sensory, and cognitive capabilities (e.g., older adults and people with impairments) throughout the entire user journey when interacting with automated buses.

This topic has only recently gained attention at AutomotiveUI (e.g., [3]) and other similar research venues. Existing studies have identified several concerns, including ingress and egress accessibility [6], emergency assistance [5], standardization of the automated bus interiors [7], user interfaces and human-machine interaction [4], and vehicle external communication [1]. Although these studies have provided valuable insights and raised awareness, their findings have not yet been fully translated into practical implementations in real public transport settings.

Building on existing studies and utilizing the *Ride the Future* platform in Linköping, we aim to bridge the gap between academic research and real-world deployment. To achieve this, we will invite

participants to co-create speculative scenarios and translate these scenarios into concrete design research questions. We expect the outcomes to have real-world impacts on the practical design and implementation of more inclusive autonomous public transportation.

2 Goals and topics

The overarching goal of the workshop is to identify and reflect on the potential safety, accessibility, and usability challenges that older adults may encounter throughout an automated bus journey. Drawing on our previously developed speculative design method [2], we adopt transportation accident investigation as a guiding metaphor for this workshop. Participants take on the role of investigators examining a fictive accident involving older adults and automated buses. The speculative accident scenario is grounded in real-world contexts and informed by official accident events and reports related to autonomous transportation. It is used to help participants construct plausible narratives to explore potential causes of the accident (e.g., passenger-related factors, autonomous systems, interaction breakdowns, and vehicle design). Specifically, the workshop aims to:

- Introduce the Ride the Future consortium and collaboratively explore opportunities for researchers and industry practitioners to use the automated bus platform in Linköping, Sweden.
- Co-speculate on scenarios that reveal unexpected and undesirable situations older adults may encounter during key user journey stages, including boarding, seating, riding, alighting, and emergency situations.
- Translate these speculative accident scenarios into design research questions that support the development of safer and more inclusive future autonomous public transport systems.

3 Expected outcomes

We expect the workshop to generate three outcomes:

- A collection of speculative scenarios that highlight critical accessibility, human-vehicle interaction, and safety challenges experienced by older adults throughout the automated bus user journey.
- A set of design research questions derived from the identified challenges, addressing the design of passenger information communication, accessibility support, emergency assistance, and other in-vehicle services. This will be achieved by bringing together researchers from diverse fields through interdisciplinary discussions.
- Reflections on the applicability of the speculative design approach for uncovering unintended consequences of autonomous public transportation. This will be achieved by applying the speculative design method that we previously developed (see [2]) and facilitating participants' reflections on the method at the end of the workshop.

We plan to summarize the expected outcomes of the workshop in a joint report or publication for AutomotiveUI or other research venues. This report or publication would serve as the foundation for forming an interdisciplinary research interest group on this topic.

4 Workshop activities and schedule

The workshop is expected to last 4 hours and consists of four stages (see Table 1). We begin with an introduction to the Ride the Future programme and its automated bus platform. This session is led by the programme's research coordinator at Linköping University. Stage 2 introduces participants to a fictive accident. We use a real operational route from a residential area to a shopping center in Linköping as the setting of the accident (see Figure 1). To support perspective-taking, participants use an empathy kit to simulate older adults' experiences. In Stages 3 and 4, participants formulate and enact their speculations about the accident, which are then used to elicit design research questions on accessibility and inclusive design challenges for the target user group. Initial questions proposed by participants during the workshop (Stage 4) are further summarized and categorized after the workshop. The finalized set of questions is shared with participants. The workshop activities are supported by 3D-printed automated bus mock-ups, LEGO, clues from the fictive accident (including video and audio scripts, images, and textual descriptions), and design prototyping tools (see Figure 1). These materials are designed to enable participants to explore external and in-vehicle design aspects, including passenger information communication, seating, and emergency support. The workshop is conducted by two facilitators and two observers.

5 Attendance and support

We welcome participants from diverse fields, including human–machine interaction, ergonomics, cognitive science, human factors, product and service design, interaction design, lighting design, accessibility and inclusive design, public transport, automated vehicle technology, and engineering. We aim for 20–35 participants. The organizers will actively recruit participants through their professional networks in academia, industry, and public sector organizations. The workshop requires a room with tables and a presentation screen, and the room should accommodate around 30 participants. All workshop materials, including empathy kits, 3D-printed automated bus mock-ups, LEGO, printouts of the fictive accident scenarios, additional prototyping and discussion materials, will be provided by the workshop organizers.

6 Biographies

Wanjun Chu is an Assistant Professor at Linköping University. His current research focuses on human–A^3 (automation, autonomous systems, and AI) interaction and how it can support the design for sustainable behavior transitions, with a particular emphasis on speculative design approaches.

Torbjörn Andersson is a Senior Lecturer at Linköping University in Sweden. He is the university's research coordinator in the Ride the Future consortium and his current research focus is on external Human–Autonomous Vehicles interaction with a focus on light design. He is the university leader for Vinnova, Tillväxtverket and Trafikverket funded research projects and is supervising several PhD students.

Fredrik Henriksson is an Assistant Professor at Linköping University in Sweden. His current research focuses on processes for sustainable transitions in the intersection between technology and human behavior.

Table 1: Workshop activities and tentative schedule

Stages	Activity descriptions	Schedule
Stage 1	Kick-off: 1). Introduction to the *Ride the Future* automated bus platform and operational contexts. 2). Introduction to the fictive accident.	45 mins
Stage 2	Explore accident scenarios and formulate speculations: 1). Participants put on an empathy kit to simulate older adults' experiences. 2). Review and discuss accident clues. 3). Formulate speculations regarding the potential causes of the accident. Break	60 mins
Stage 3	Enact speculative scenarios and discuss design challenges: 1). Participants present their speculations through enactments of the accident scenario. 2). Discuss potential contributing factors and design elements that may have led to the accident. Break	45 mins
Stage 4	Reflection and future collaboration: 1). Identify similar interaction scenarios that older adults may encounter throughout an automated bus journey. 2). Discuss design research questions that need to be addressed to improve accessibility, safety, and usability in autonomous public transportation. 3). Explore opportunities for future work and collaboration.	60 mins



Figure 1: (Left) Automated bus operating in a residential area in Linköping, Sweden. (Right) Materials to be used in the workshop.

Simon Schütte is a Senior Associate Professor for Product Design and Affective Engineering at Linköping University. He has a background in mechanical engineering and human factors. His main research focuses on customer tolerance and acceptance.

Elena Jiménez Romanillos is a PhD student at Linköping University exploring how consumers navigate traditional grocery systems to understand the real environmental impact of everyday shopping habits. Her research also includes design fiction and speculative design as methods.

Franziska Babel is an Assistant Professor for Cognitive Science at Linköping University in Sweden. She has a background in psychology, human factors and human-robot interaction. Her research

interests are amongst others on studying acceptance and trust in autonomous systems by pedestrians and cyclists in public robot deployments. She focuses on mixed-methods and interaction design for public robots.

References

[1] Torbjörn Andersson and Fredrik Henriksson. 2025. Not Throwing Cyclists Under the (Autonomous) Bus. In *HCI International 2025 Posters*, 2025. Springer Nature Switzerland, Cham, 307–313. https://doi.org/10.1007/978-3-031-94159-7_32

[2] Johan Blomkvist, Wanjun Chu, and Emma Mainza Chilufya. 2025. Sofia Fled or Died? Design Fictional Explorations of Unintended and Unsustainable Consequences of Human-AI Interaction. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (*CHI EA '25*), April 25, 2025. Association for Computing Machinery, New York, NY, USA, 1–10. https://doi.org/10.1145/3706599.3719936

[3] Henrik Detjen, Stefan Geisler, Stefan Schneegass, Andrew L Kun, and Vidya Sundar. 2021. Workshop on the Design of Inclusive and Accessible Future Mobility. In *13th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (*AutomotiveUI '21 Adjunct*), September 22, 2021. Association for Computing Machinery, New York, NY, USA, 194–196. https://doi.org/10.1145/3473682.3479719

[4] Julian Faig, Gregory-Jamie Tüzün, Daniel Roth, and Matthias Kreimeyer. 2025. Challenges experienced by disabled people when traveling on and interacting with an autonomous bus. *Proceedings of the Design Society* 5, (August 2025), 891–900. https://doi.org/10.1017/pds.2025.10103

[5] Aaron Gluck, Kwajo Boateng, Earl W. Huff Jr., and Julian Brinkley. 2020. Putting Older Adults in the Driver Seat: Using User Enactment to Explore the Design of a Shared Autonomous Vehicle. In *12th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (*AutomotiveUI '20*), September 20, 2020. Association for Computing Machinery, New York, NY, USA, 291–300. https://doi.org/10.1145/3409120.3410645

[6] Nade Liang, Ashima Krishna, Brandon J. Pitts, Stacey Connaughton, Ksenia Kozak, and Konstantina Gkritza. 2025. Understanding older adults' needs for and perceptions of shared autonomous vehicle interior features: A focus group and user enactment study. *Applied Ergonomics* 123, (February 2025), 104408. https://doi.org/10.1016/j.apergo.2024.104408

[7] Robin Severs, Jiayu Wu, Cyriel Diels, Dale Harrow, Joseph Singleton, and Richard Winsor. 2022. Imagining an Inclusive Future for Shared Autonomous Vehicle Interiors: A Participatory Design Workshop Study. In *Adjunct Proceedings of the 14th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (*AutomotiveUI '22*), September 17, 2022. Association for Computing Machinery, New York, NY, USA, 73–78. https://doi.org/10.1145/3544999.3554787